

SIGN BRIDGE

A Real-time Bidirectional Ugandan Sign Language Interpretation
System

Engineering And Research Proposal



Different Hands Same Voice

BSE27-30

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Abstract

SignBridge is a proposed real-time, bidirectional Ugandan Sign Language (USL) interpretation system that connects Deaf USL signers and hearing non-signers through two directions of communication: Sign $\rightarrow$ Speech and Voice $\rightarrow$ Sign. The project treats interpretation as an engineering lifecycle rather than as a classification problem alone. The lifecycle begins with ethically governed, standardized and deliberately varied video collection. It continues through provenance-aware storage, technical review, linguistic validation, landmark extraction, model training and signer-aware evaluation, and it ends in a cross-platform Flutter application supported by a FastAPI service.

The first data target is a corpus of 50 structured sentences. The prototype does not seek volume. It seeks a rich dataset, in which each sentence is recorded under a planned combination of capture format, orientation, background and lighting, so that the value of the dataset comes from measured variation rather than from repeated, near-identical recordings. This is a proof-of-engineering-scale dataset and not a claim of national USL coverage. PostgreSQL stores metadata while the video itself is kept in separate file and object storage, so the architecture can grow well beyond the prototype without redesign. The system is online only, and offline operation is outside the scope of this project.

The literature shows that Ugandan learning resources, dictionaries, interpreter-service platforms, academic translation systems and national digitization initiatives already exist. SignBridge therefore does not claim that USL technology is new. Its proposed engineering contribution is the integration of controlled data collection, validation, large-video storage, cloud upload and persistence, dataset versioning, confidence-aware inference and cross-platform deployment into one reproducible workflow.

Project Summary

Table 1 summarizes the prototype at a glance. The remaining sections define each element in detail.

Table 1: Prototype parameters

Item	Definition
Prototype corpus	50 structured sentences, each a registered entry in the Sign Registry
Primary recordings	150 recordings (50 sentences $\times$ 3 capture formats), plus retakes where quality review or validation requires them
Sources of variation	Signer, capture format and orientation, lighting, background, capture device, distance and signing speed
Collection community	Makerere University Deaf Students' Association (MUDSA)
Client	Flutter, on Android and Web/PC
Backend	FastAPI with PostgreSQL for metadata and separate cloud storage for video
Vision and classification	MediaPipe landmarks and a sentence-level USL classifier
Runtime	Sign $\rightarrow$ Speech and Voice $\rightarrow$ Sign
Connectivity	Online only; offline operation is not part of this project
Core contribution	A traceable, reproducible and testable pipeline from raw USL data to deployment

1. Background and Motivation

Ugandan Sign Language is a language-specific visual-gestural system. A useful automated interpreter must therefore do more than detect hands. It has to learn approved signs from real signers, preserve the variation between them, distinguish linguistic correctness from camera quality, communicate uncertainty, and run on hardware that users already own.

The project defines two runtime pipelines. Sign $\rightarrow$ Speech takes camera input, extracts hand landmarks, classifies the performed USL sentence, applies context and language handling, and produces speech. Voice $\rightarrow$ Sign takes speech, converts it to text, maps the text to the approved vocabulary and context, and presents a sign representation. In normal operation the recognition path sends compact landmark sequences to the server rather than continuously uploading raw video.

Raw video nevertheless remains essential during dataset construction. It is the auditable source from which landmarks and any future features can be regenerated, and it allows reviewers and validators to inspect exactly what the signer performed.

2. Problem Statement

Existing USL-related tools demonstrate important capabilities, but they address different parts of the problem. Learning applications provide reference content, interpreter platforms connect people to human interpreters, academic projects demonstrate recognition or translation models, and dictionary and national digitization initiatives develop language resources. The engineering gap addressed by SignBridge is the controlled connection of these lifecycle stages.

The specific problem is how to engineer a USL interpretation system whose training data are ethically collected, consistently labeled, varied enough to test generalization, traceable from raw video to model input, stored safely at video scale, and connected to a real-time application with explicit uncertainty handling. The problem is sharpened by the constraint of a small, bounded dataset. With a prototype corpus of 50 sentences, the usefulness of the data depends on how deliberately it is varied rather than on how many recordings it contains.

3. Aim and Objectives

3.1 Aim

The aim of the project is to design, implement and evaluate a cross-platform, real-time bidirectional USL interpretation system supported by a reproducible data-collection, validation and machine-learning pipeline.

3.2 Objectives

To achieve this aim, the project has the following objectives.

1. Design a cloud-connected SignBridge Data Collector for standardized USL video recording, guided capture, metadata capture, review and validation.
2. Collect a prototype dataset covering 50 structured sentences, each recorded in three guided capture formats under planned variation in background, lighting and orientation. This yields 150 primary recordings, with additional recordings wherever quality review or validation requires a retake.
3. Capture signer, capture format, orientation or sweep state, lighting, background, capture device, time, validator and validation status as structured metadata.
4. Store raw videos in scalable file and object storage while PostgreSQL holds metadata, provenance, lifecycle and validation records.
5. Extract MediaPipe landmark representations and establish a reproducible preprocessing pipeline.
6. Train and evaluate suitable USL sentence classifiers, using signer-aware and condition-held-out evaluation where the sample structure allows it.
7. Deploy recognition behind FastAPI and integrate it with the Flutter Android and Web/PC clients.
8. Implement confidence-aware behavior that recognizes a sentence when sufficiently confident, asks for clarification when uncertain, and reports unsupported or no-match cases rather than inventing a class.
9. Implement Voice → Sign through speech-to-text, vocabulary and context mapping, and a sign representation layer.
10. Evaluate the system at the data, model, API, application, usability and operational levels.

4. Research Questions

The project is guided by the following research questions.

1. How does a standardized, metadata-driven recording protocol affect the consistency and traceability of a prototype USL dataset?
2. How well do MediaPipe landmark representations support recognition of the 50 selected structured sentences under controlled but varied conditions?
3. How does signer-aware evaluation change measured performance compared with a random split?
4. How sensitive is recognition to each planned source of variation when lighting, background or capture format is held out of training?
5. What confidence policy balances useful recognition against uncertain outputs?
6. Can the same collection workflow operate reliably on Android and Web/PC while preserving metadata and video integrity?
7. What engineering controls are required to maintain a reproducible relationship between raw video, landmarks, dataset versions and deployed models?

5. Scope and Boundaries

The prototype corpus consists of 50 structured sentences taken from the project's source-of-truth instrument. No sentence is invented during collection, and each is validated by a fluent signer or interpreter before it enters the Sign Registry. Recognition operates at the level of these registered sentences. Each sentence is a class, and the system does not attempt open-ended continuous translation.

The prototype dataset comprises 150 primary recordings, made up of 50 sentences in each of three capture formats, with lighting and background varied by a balanced rotation described in Section 9. The architecture is designed to grow beyond this size without changing the separation between metadata, video assets, model artifacts and runtime services.

Android and Web/PC are first-class client platforms, and the PC client supports both built-in and USB webcams. Video assets are stored in Firebase Cloud Storage, while PostgreSQL holds metadata, provenance and validation records. The runtime covers both Sign → Speech and Voice → Sign.

The system is online only. Both the collector and the runtime require an active connection to the FastAPI service. Local offline storage and synchronization are outside the scope of this project.

The project does not claim national-scale coverage, universal USL understanding, open-ended sentence translation, or replacement of professional interpreters.

6. Literature Review

The review covers Uganda-specific systems, African sign-language datasets and the methodological evidence relevant to dataset design. Table 2 summarizes what each source describes and the engineering gap SignBridge proposes to address.

6.1 Summary of Reviewed Literature

Table 2: Literature reviewed and the gaps SignBridge addresses

Source / System	Key Contribution	Gap or Limitation	SignBridge's Response
Usign (UNAD & Javanet Systems) – Ugandan learning app	Alphabet, numbers, time, food and a first set of 100 words	Reference content, not machine-learning-ready training data	Builds a versioned Sign Registry with structured, ML-ready collection
DeafAccess – Ugandan interpreter platform	Connects users with human interpreters	Not automated recognition	Offers automated recognition for a bounded sentence set, not a replacement for interpreters
Makerere CoCIS Digital USL Dictionary	Foundational USL lexical resource	Reference resource, not structured for training	Draws vocabulary context from it while adding collection, validation and runtime capability
UNAD & SPIDER digitization initiative	Digitalising USL content for social media, YouTube, eLearning, mobile and DVD	Content digitization, not an engineered ML pipeline	Complements this work with a data-to-deployment engineering lifecycle
UCC / UCUSAF digitization initiative	UGSL digitization, standardization and capacity building	Concerns national resources and standards, not a bounded prototype	Produces traceable engineering evidence compatible with this broader effort
Nakuwanda et al. (2024), Makerere	Real-time USL-to-speech system using MediaPipe, hand/pose tracking and sequence models	Model-centric; the data lifecycle around it is not engineered	Engineers the upstream data lifecycle and downstream deployment around similar recognition components
Suubi et al. (2024), Makerere	ML mobile app with bidirectional conversion, gesture capture and Firebase	Collection and versioning are not formalized	Adds database-backed recording identifiers, storage keys and checksums
Katende et al. (2024), Makerere	Deep-learning sign-to-text system with SHAP, Grad-CAM and LIME explainability	Focused on explainability rather than data traceability	Complements with engineering traceability from raw video to deployed model
Batte (2024), review	Survey of visual, glove-based, avatar and mobile translation approaches	Confirms camera position and lighting as recognition factors	Treats orientation, lighting and background as controlled, recorded metadata

Source / System	Key Contribution	Gap or Limitation	SignBridge's Response
CASL-W60 dataset	60-word Central African dataset from 19 volunteers, with MediaPipe landmarks and signer-aware splitting	—	Supports SignBridge's use of signer-aware splitting and metadata/landmark separation
FluentSigners-50 dataset	Signer-independent evaluation with variation in posture, distance, lighting, background, angle and device	Random splitting can leak signer characteristics into test data	Motivates deliberate variation design and signer-aware evaluation
AfriSign dataset	Multi-country video-to-text African sign-language dataset	African sign-language data remains scarce	Reinforces the value of a traceable, reproducible Ugandan dataset
Ghanaian sign-language dataset (2026)	9,879 video samples totaling 23.15 GB	Video at this scale strains relational storage	Keeps metadata in PostgreSQL and video in separate file/object storage

6.2 Literature Gap and Positioning

Two methodological decisions follow directly from the table above. First, variation (orientation, signer, distance, speed, lighting, background) must be treated as a measurable data attribute rather than an incidental property of recordings, as the African dataset literature demonstrates. Second, video-scale storage must be engineered as an infrastructure concern separate from relational metadata, as the Ghanaian dataset's 23.15 GB illustrates.

Across the Ugandan and academic sources, the recurring gap is that reference content, translation prototypes and ad hoc collection practices do not, on their own, add up to a traceable, reproducible engineering lifecycle. SignBridge's proposed contribution is that lifecycle: a versioned Sign Registry, database-backed provenance, signer-aware evaluation, confidence-aware inference, cross-platform first-class clients, and separated metadata/video storage, connected end to end rather than addressed piecemeal.

7. Contribution and Positioning

7.1 What SignBridge Improves

The defensible contribution is not that each individual component is novel; camera-based recognition, MediaPipe landmarks, mobile applications, dictionaries and bidirectional translation all have precedents. The following fourteen aspects describe where SignBridge's engineering integration and controls add value across the machine-learning lifecycle.

1. Controlled Data Collection

Uses a standardized 3-format guided capture protocol instead of many separate orientation recordings.

- ❖ Captures horizontal sweep, vertical sweep, and natural signing.

- ❖ The 45° target introduces controlled orientation variation without making collection unnecessarily complicated.

2. Data Variation

Deliberately captures variation in orientation, signer, distance, signing speed, lighting, and background.

- ❖ This is important because the model should not only work under one fixed recording condition.

3. Data Traceability and Provenance

Every recording is connected to structured metadata.

- ❖ Recording IDs, signer information, sign, capture format, conditions, validation and storage references can be traced together.

4. Separation of Video and Metadata

PostgreSQL stores structured metadata rather than large video files.

- ❖ Actual videos are stored in dedicated cloud/object storage.
- ❖ This makes the architecture suitable for datasets that grow beyond the initial prototype.

5. Community and Linguistic Validation

Separates technical video quality from linguistic correctness.

- ❖ A video can be technically good but still contain an incorrect or unclear sign.
- ❖ SignBridge therefore provides separate quality review and linguistic validation.

6. Dataset Versioning

Records can be included in controlled dataset versions.

- ❖ This creates a reproducible relationship between the data used for training and the resulting model.

7. Machine-Learning Reproducibility

Raw videos remain available as the source from which landmarks and derived features can be regenerated.

- ❖ Preprocessing and model versions can therefore be tracked.

8. Signer-Aware Evaluation

Where the available signer structure permits it, SignBridge evaluates models using signer-aware train/validation/test separation rather than relying only on random splitting.

- ❖ This helps investigate whether performance generalizes beyond people already represented in training data.

9. Confidence-Aware Recognition

The system does not have to force every input into a known sign.

- ❖ It can recognize, request clarification/repetition, or report no-match/unsupported cases depending on confidence.

10. Cross-Platform Collection

Android and Web/PC are treated as first-class clients.

- ❖ A phone can be used for collection, while a PC can use a built-in or USB webcam.
- ❖ The same collection workflow and backend contracts are maintained across platforms.

11. Cloud-Based Accessibility

The system is designed for hosted access rather than depending on one researcher's computer.

- ❖ Multiple authorized users can access the collection system through the deployed application.

11. End-to-End Engineering Integration

SignBridge connects sign registry to consent to guided recording to cloud storage to validation to dataset to landmark extraction to ML to API then to the application, rather than treating the classifier as the entire project.

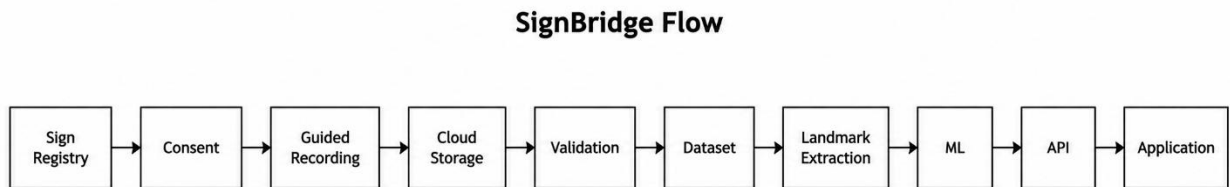


Figure 1: sign-bridge integration flow

13. Security and Governance

Participant consent, role-based access, audit trails, controlled identifiers, retention/deletion procedures and protection of raw videos are incorporated into the engineering lifecycle.

14. Operational Testing

Testing is not limited to ML accuracy.

- ❖ SignBridge also tests the camera, upload/storage, API, database, UI, security, recovery, performance and cross-platform workflow.

In short:

SignBridge improves the complete USL interpretation engineering lifecycle—from controlled and traceable data collection, through cloud storage and linguistic validation, to reproducible machine learning, confidence-aware recognition, and cross-platform deployment.

This is consistent with the proposal's existing framing that the contribution is the integration and engineering controls around the ML lifecycle, rather than claiming that individual components such as MediaPipe, camera recognition, or sign-language classification are themselves new.

7.2 Relationship to Existing Ugandan Tools

Usign serves learning and reference, whereas SignBridge adds real-time recognition and an engineered dataset lifecycle.

DeafAccess provides interpreter booking and communication services, whereas SignBridge offers automated recognition for a bounded set of sentences and is not a replacement for human interpreters.

The digital USL dictionary is a lexical and reference foundation from which SignBridge can draw vocabulary context, while SignBridge adds collection, validation and runtime capabilities.

Academic USL translators demonstrate recognition and translation research, and SignBridge adds storage, provenance, dataset versioning and operational quality assurance around that work. The national UGSL digitization effort concerns resources, standardization and digital inclusion, and SignBridge is a bounded research prototype that produces traceable engineering evidence compatible with that broader need.

8. System Overview and End-to-End Workflow

The complete workflow runs from the definition of the approved corpus to the release of a deployed, documented system. It is described here in five stages.

8.1 Preparation

The team first defines the approved sentence corpus, the recording protocol, the ethics requirements and the evaluation criteria. For each participant, a signer is registered and consent is confirmed. A collection session is then created that records the venue, the protocol version and the device context.

8.2 Collection

The collector selects an approved sentence and a guided capture format, and the system applies the corresponding guided sweep protocol automatically. The collector confirms the lighting and background and selects the camera. On Android this is the phone camera, and on Web/PC the system diagnoses the built-in or USB webcam. The collector records the sentence and previews the take. Once the take is accepted, the video is uploaded to cloud storage, and it is not treated as complete until its integrity has been verified. If an upload is interrupted, the collector is prompted to retry.

8.3 Review and Validation

Each uploaded recording undergoes technical quality review, and a failed take is retaken. A fluent signer, an interpreter or a designated validator then performs linguistic validation. Records that satisfy every inclusion condition are frozen into a versioned dataset.

8.4 Model Development

MediaPipe landmark extraction and preprocessing are run on the frozen dataset version. Baseline and candidate classifiers are trained and evaluated on validation and test partitions using signer-aware and condition-held-out logic where feasible. The selected model is packaged together with its preprocessing configuration and dataset version.

8.5 Deployment and Release

The model is deployed behind FastAPI. At runtime, Sign → Speech flows from camera to landmarks, classifier, confidence policy, context and language layer, and speech. Voice → Sign flows from microphone to speech-to-text, vocabulary and context mapping, and a sign representation. The team then runs end-to-end, cross-platform, security, recovery and usability evaluation, and releases the versioned application together with dataset documentation, model documentation, and backup and recovery artifacts.

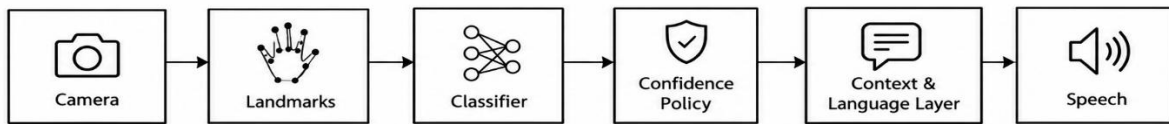
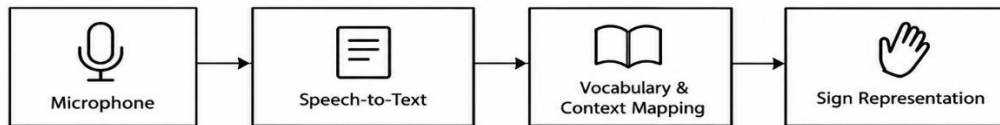
Sign → Speech (at runtime)**Voice → Sign (at runtime)**

Figure 2: shows sign to speech at run time and voice to sign at run time

9. Data Collection Design

9.1 The Prototype Corpus

The first data target is a corpus of 50 structured sentences. A structured sentence is a short, fixed-form USL sentence drawn from the approved source-of-truth instrument and registered in the Sign Registry with a stable identifier, an approved English text and a validation record. Where the instrument groups sentence into functional categories, those categories are recorded so that coverage can be reported by category.

Working at sentence level means that every recording is a complete and meaningful utterance. This is closer to how the system will be used than isolated signs, and it avoids the need to segment continuous signing in the prototype.

9.2 Variation over Volume

The prototype deliberately favors variation over volume. A dataset built from many near-identical recordings of a small number of sentences would give an inflated impression of robustness, because the model would be tested on conditions it had already seen. SignBridge instead fixes the number of sentences at 50 and spends its recording effort on varying the conditions under which each sentence is captured. These conditions are the format and orientation of the capture, the background, the lighting, the signer and the capture device. Distance and signing speed are allowed to vary naturally within collector guidelines and are recorded where they can be measured.

Each dimension of variation is controlled and recorded as structured metadata rather than free text. The metadata can therefore be used later to explain model behavior, to build evaluation partitions, and to report honestly which conditions the dataset does and does not cover.

9.3 Capture Formats

Every sentence is recorded in three guided capture formats. In the horizontal sweep format, the signer performs the sentence during a continuous left-to-right sweep of orientation relative to the camera, moving from the center to approximately 45° left, back to the center, to approximately 45° right, and back to the center. In the vertical sweep format, the sweep runs from the center to an upward orientation, back to the center, to a downward orientation, and back to the center. In the natural performance format, the signer performs the sentence at a comfortable, normal orientation without a forced sweep.

The sweep formats deliberately expose the classifier to the changes in apparent hand shape and position that occur when the signer is not squarely facing the camera. The natural performance format provides the reference condition closest to real use. The collector does not set the orientation by hand. Selecting a capture format applies the corresponding guided protocol automatically, so that every signer is recorded under the same procedure.

9.4 Environmental Controls

Lighting is limited to three controlled values, DAYLIGHT, INDOOR_ARTIFICIAL and LOW_LIGHT_DIM. Background is limited to three controlled values, PLAIN_WALL_LIGHT, PLAIN_WALL_DARK and TEXTURED_CLUTTERED. Together they define nine environment pairings. The venue is Makerere University and MUDSA, and protocol version v1 is recorded on every collection session. Values are chosen from controlled lists and are never typed, which prevents spelling variants and keeps the metadata directly usable for analysis.

9.5 Signers and Community Engagement

Recordings are collected through the Makerere University Deaf Students' Association, whose members are fluent USL users. Informed consent is obtained before any recording, each signer is identified by a controlled signer identifier rather than a name, and each sentence is validated by a fluent signer or interpreter.

Signer variation is the most valuable single source of variation in a small dataset, because differences in signing style, hand size, speed and habit cannot be reproduced by changing the camera. The plan therefore targets at least three signers, which is the minimum needed to hold one signer out for evaluation while still training on the others, and it recruits further signers as MUDSA availability permits. The number of signers actually achieved will be reported as a limitation of the dataset, whatever it turns out to be.

9.6 Allocation of Variation across the Corpus

Table 3 lists the planned dimensions of variation and how each is introduced.

Table 3: Dimensions of planned variation

Dimension	Values	How it is varied	Recorded as
Capture format and orientation	Horizontal sweep; vertical sweep; natural performance	Every sentence is recorded once in each format	Capture format and sweep state
Lighting	Daylight; indoor artificial; low light or dim	Assigned by rotation across sentences and formats	Controlled list value
Background	Plain wall (light); plain wall (dark); textured or cluttered	Assigned by rotation together with lighting	Controlled list value
Signer	Three or more MUDSA signers	Recordings distributed across signers by rotation	Controlled signer identifier
Capture device	Android camera; built-in webcam; USB webcam	Varies by session according to the device in use	Device and camera context
Distance and speed	Natural variation within guidelines	Observed and not forced	Recorded where measurable

The three capture formats, three lighting conditions and three backgrounds could in principle be combined in 27 ways for every sentence. Recording every combination would require 1,350 recordings and would defeat the purpose of a bounded prototype. The plan instead uses a balanced rotation. Each sentence is recorded once in each of the three capture formats, giving 150 primary recordings, and the three recordings of a sentence are placed in three different environment pairings. For sentence number s from 1 to 50 and capture format number f equal to 0, 1 or 2, the pairing used is number $(s + 3f)$ modulo 9, counted over the nine pairings in the protocol. Under this rule every pairing is used for five or six sentences in each format and for 16 or 17 recordings overall, no sentence is confined to a single environment, and no environment is confined to a single format or group of sentences.

This balance matters for the classifier as much as for coverage. If a group of sentences were always recorded against the same background, a model could learn the background instead of the sign. Rotating the pairings prevents that association and allows the evaluation to treat lighting and background as factors that can be held out.

In practice, sessions are organized by environment. Each of the nine pairings is set up once, and all recordings assigned to that pairing, between 16 and 17 in total, are captured in that session. Recordings are distributed across the participating signers by rotation so that each signer is recorded in all three capture formats where attendance permits. A retake that replaces a failed recording keeps the same cell of the plan. Retakes and any voluntary additional passes by signers are stored as additional recordings and never overwrite earlier ones.

9.7 Identifiers and File Naming

Each sentence has a sentence identifier, and each recording receives a unique recording identifier generated by the server. Video files are named using the pattern `SignerID_SentenceID_Format_Take`, where the take number defaults to 01 and increases for retakes. The file extension reflects the container that the browser or device actually produced, whether webm or mp4, and the true container and MIME type are recorded in the metadata. Lighting, background and other conditions are kept in metadata and not in the file name, so that names remain short and unambiguous.

10. Data Quality and Validation

A technically good video is not automatically a linguistically correct sample, so SignBridge makes two separate decisions about every recording.

The first is technical quality. The reviewer checks that the signer is fully visible, that both hands are visible where required, that the complete sentence was performed without interruption, and that the lighting and background match the values recorded for the take. A take can pass, be retaken, or be excluded. The second is linguistic correctness, which is judged by a fluent signer or interpreter as correct, unclear or incorrect. Because the recordings are complete sentences, the validator confirms both that the full sentence was performed and that its signs appear in the intended order.

Three further checks complete the picture. Consent must be authorized, metadata must be complete, and the stored file must pass checksum verification. A recording is included in a dataset only when consent is authorized, technical quality has passed, the metadata are complete, the sentence has been validated as correct, and the checksum has been verified. An unclear validation places the recording under review, and an incorrect one excludes it. Excluded recordings are retained with the reason for exclusion, because the pattern of rejections is itself useful information about the collection process.

11. Video Storage

Video bytes are not stored as large binary objects in PostgreSQL. PostgreSQL stores recording metadata and the storage key, and a separate file and object storage layer holds the videos. The deployed architecture uses Firebase Cloud Storage for raw video assets, and Firebase Hosting provides the hosted web client. In the running system, the Android and Web clients communicate with FastAPI, which writes metadata to PostgreSQL and video to the object store.

Landmark artifacts are held in dataset and artifact storage, models are held in versioned model storage, and dataset exports are held in controlled backup storage. Uploads travel over HTTPS with retry and integrity verification, and a recording is not marked complete until the stored file

has been verified. This separation keeps PostgreSQL focused on structured metadata and allows video collections to grow to tens of gigabytes without redesign.

12. Machine-Learning Pipeline

12.1 Data Preparation and Feature Extraction

The pipeline ingests only records that belong to a declared dataset version. It verifies video integrity and metadata completeness, decodes the video and samples frames consistently, and extracts MediaPipe hand and pose landmarks. Raw landmark outputs and derived features are stored separately. Coordinates are then normalized and organized into temporal sequences that cover the whole sentence, using a declared preprocessing version so that every result can be reproduced.

12.2 Model Development

A baseline model is trained before any more complex model so that gains from added complexity can be measured. Candidate models are then compared against the baseline. Because the corpus is small, landmark-level augmentation may be used during training, and augmented samples are generated only from training partitions so that no derived copy of a test recording reaches the training data. The selected model is packaged with its preprocessing configuration and dataset lineage and served through FastAPI, so that every deployed model can be traced to the data that produced it.

12.3 Evaluation Strategy

With 150 primary recordings, each sentence has three recordings. The evaluation therefore emphasizes controlled comparisons and not a single headline accuracy. Where the signer structure permits, signer-aware evaluation holds out all recordings of one signer and trains on the others. The variation design also allows condition-held-out evaluation, in which all recordings made under one lighting condition, one background or one capture format are held out, so that the sensitivity of the model to each factor can be measured directly. A random split is reported alongside these for comparison, so that the effect of same-signer leakage is visible.

Because fixed training, validation and test partitions would leave some sentences unrepresented in one of the partitions, grouped cross-validation by signer or by condition is used. The metrics reported are accuracy, macro-F1, per-class recall, the confusion matrix and inference latency. Results from a dataset of this size are reported as prototype measurements with their limitations and are not presented as evidence of general accuracy.

12.4 Confidence Policy

Confidence is treated as an engineering signal and not hidden. Thresholds are tuned on validation data, and the resulting behavior is summarized in Table 4.

Table 4: Confidence policy

State	Behavior
High confidence	Return the recognized sentence.
Medium confidence	Ask the user for clarification or to repeat the sentence.
Low confidence	Report that the sentence could not be recognized reliably.
Unsupported or no match	Do not invent a sentence and do not silently map to another class.

13. Bidirectional Runtime

13.1 Sign → Speech

The camera stream is processed on the client to extract MediaPipe landmarks. Compact landmark sequences covering the sentence being signed are sent to the FastAPI service, where the classifier returns a candidate sentence and a confidence value. The confidence policy decides whether the result is presented, a clarification is requested, or the attempt is reported as unrecognized. A recognized sentence passes through the context and language layer, which in the prototype maps it to its approved text and validates the output, and it is then delivered as text and speech. Raw video is not the normal runtime payload, because compact landmark sequences reduce latency and bandwidth.

13.2 Voice → Sign

The microphone input is converted to text by speech-to-text. The text is matched against the registered sentences and their context, and the matched sentence is presented through the sign representation layer, which may be an approved recorded performance or an animation. If the spoken concept lies outside the supported corpus, the system states that limitation rather than fabricating a sign.

14. Cross-Platform Engineering

Android and Web/PC are first-class platforms that share Flutter business logic and API contracts. Platform capabilities are reached through abstractions. A camera service provides the phone camera on Android and the built-in or USB webcam in the browser, a video storage service handles upload to cloud storage, an HTTP client abstraction manages HTTPS communication, and

repository interfaces isolate the application from the API. The key rule is that feature code must not depend directly on Android-only camera, filesystem or secure-storage interfaces.

The application requires connectivity and does not implement local offline storage or synchronization.

15. Database and API Architecture

The database holds the following core entities: users, signers, consents, signs, sign versions, collection sessions, recordings, recording conditions, videos, recording reviews, sign validations, dataset versions, dataset members, exports and audit logs. The signs table holds the registered sentences, and sign versions preserve the meaning of historical data if an approved sentence is later revised.

The representative API covers authentication and token refresh, signers, signs, sessions, recordings, video upload, review, validation, coverage, exports and health checks. Each recording moves through an explicit lifecycle: created, recording, uploaded, review pending, then quality passed or quality failed, validation pending, and finally included, review required or excluded before it is assigned to a dataset version.

16. Testing and Evaluation

Testing is organized by level. Backend unit and API tests cover business rules, permissions, the recording lifecycle, upload, review, validation and export. Database tests cover migrations, foreign keys, uniqueness and constraints. Flutter unit and widget tests cover validators, state handling and all major interface states, and integration tests exercise the complete collection workflow.

Physical device testing verifies Android camera capture, video recording, upload, and recovery from an interrupted upload. Web testing verifies Chrome camera permissions and the built-in and USB webcams. Security testing covers unauthorized access, role escalation, malicious uploads and file type and size limits. Performance testing covers pagination and representative video uploads, and recovery testing covers database and video restore with checksum verification.

Machine-learning testing covers the partitioning logic, the metrics, the confidence thresholds and inference latency. Visual and usability testing covers touch targets, responsive layout and workflow usability, with issue logs and with participation from Deaf and USL community members.

17. Ethics and Data Governance

Informed consent is obtained before any recording. The consent scope states that recordings may be stored, used to train and evaluate models and, where applicable, shown in the application as a

sign representation, and a signer may withdraw consent, in which case the affected recordings are excluded. Controlled signer identifiers are used in file names and research exports wherever possible, and access to raw video is restricted by role.

An audit trail is maintained for validation and dataset inclusion, and retention, deletion and backup procedures are defined. Participant video is not published without appropriate consent and governance. Deaf and USL community members take part in validation and in usability evaluation, and the project reports its vocabulary, signer and sample-size limitations honestly.

18. Implementation Roadmap

The work begins with a foundation phase in which the repository, architecture, protocol and brand are locked. The collector core follows, comprising a responsive Flutter shell and design system, authentication and role-based access, signer registration and consent, the Sign Registry and protocol, and collection sessions. The capture phase then adds the platform service abstractions, recording setup with controlled conditions, Android camera recording, and Web camera and USB webcam recording.

Review and validation, the dashboard, the records view and coverage reporting come next, followed by cloud upload and persistence and then dataset export, backup and versioning. A hardening phase covers security, performance, recovery, cross-platform regression and visual quality assurance. Once the dataset has been collected, landmark extraction, model training, evaluation and deployment behind FastAPI proceed on the frozen dataset version. The project concludes with a browser demonstration of the interface, Android field acceptance, and release and handover.

19. Final Acceptance Flow

Acceptance begins by starting PostgreSQL and FastAPI and verifying their health, then launching Flutter Web in Chrome and logging in as a data collector. The tester registers or selects a signer, verifies consent, creates a session, and selects an approved sentence and capture format. The tester verifies that the correct protocol preset is applied automatically, opens the camera, selects the built-in or USB webcam, and records, stops and verifies the actual video asset.

The tester then reviews quality and validates the sentence, uploads the recording, and verifies both the PostgreSQL metadata and the Firebase Cloud Storage video asset. The browser is refreshed to verify persistence, and an interrupted upload is tested for retry and recovery. Cloud persistence and file integrity are verified, a dataset export is generated and its metadata compared against the videos, and the retained automated test suite is run. Finally, the critical path is repeated on Android.

20. Claims and Research Discipline

The project may state that its prototype target is 50 structured sentences recorded under planned variation, but it must not claim that this is enough for production-scale USL translation. It may state that the architecture supports large video collections through separate video storage, but it must not claim that PostgreSQL is the preferred primary store for multi-gigabyte video.

It may state that signer-aware evaluation is used where feasible, but it must not claim that high random-split accuracy proves real-world generalization. It may state that confidence-aware behavior reduces forced predictions, but it must not claim that the system always understands every USL sign or sentence. Finally, it may state that its contribution is an integrated engineering pipeline, but it must not claim that every individual component is novel.

21. Conclusion

SignBridge should be presented as a complete software-engineering and machine-learning system, and not as a claim that sign-language recognition itself was invented by this project. Existing Ugandan work establishes the importance of dictionaries, learning resources, interpreter services, USL recognition and digitization. The engineering value proposed here is the controlled bridge from community-validated USL data to a tested, deployable and uncertainty-aware interpretation service.

The prototype is intentionally bounded, at 50 structured sentences recorded under deliberately planned variation, so that the team can finish a measurable system and learn from it. The architecture is scalable. More approved sentences, signers, recordings, object storage, model versions, validation and external collaboration can be added without changing the fundamental separation between metadata, video assets, model artifacts and runtime services.

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Internal Project Sources

SignBridge Product and Architecture Overview: Real-Time Ugandan Sign Language Interpreter.

SignBridge Data Plan and Data Collection Instrument: MUDSA engagement, consent and controlled recording.

SignBridge Data Collection Tool Master Full-Implementation Plan: cross-platform collector, PostgreSQL metadata, separate video storage, cloud upload and persistence, dataset export and test strategy.